2.2 Prokaryotic and Eukaryotic Cells

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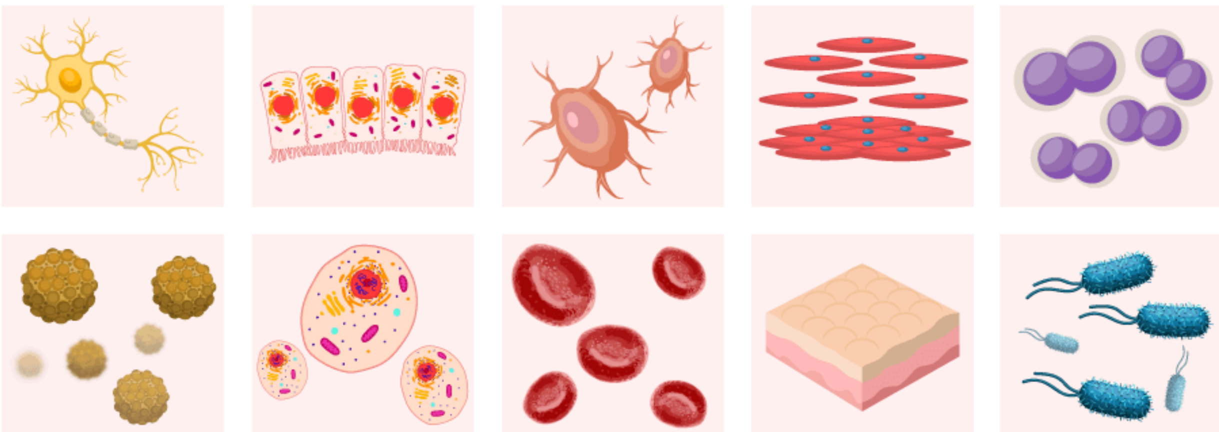
What you will learn

- The two types of cells: prokaryotic and eukaryotic
- How **eukaryotic cells** differ in structure from **prokaryotic cells**



[Figure 1]

How many different types of cells are there?

There are many different types of cells. For example, in you there are blood cells and skin cells and bone cells, and even bacteria. Here we have drawings of bacteria and animal cells. Can you tell which depicts various types of bacteria? However, all cells - whether from bacteria, human, or any other organism - will be of two general types. In fact, all cells other than bacteria will be one type, and bacterial cells will be other. And it all depends on how the cell stores its DNA.

There is a basic cell structure that is present in many but not all living cells: the **nucleus**. The **nucleus** of a cell is a structure in the cytoplasm that is surrounded by a membrane (the nuclear membrane) and contains, and protects, most of the cell's DNA. Based on whether they have a nucleus, there are two basic types of cells: prokaryotic cells and eukaryotic cells.

Prokaryotic vs. Eukaryotic Cells (Updated)
Amoeba Sisters

Watch on

Eukaryotic Cells

Eukaryotic cells are cells that contain a nucleus. Eukaryotic cells are usually larger than prokaryotic cells, and they are found mainly in multicellular organisms. Organisms with eukaryotic cells are called **eukaryotes**, and include fungi, animals, protists, and plants.

Eukaryotic cells also contain other **organelles** besides the nucleus. An **organelle** is a structure within the cytoplasm that performs a specific job in the cell. Organelles called mitochondria, for example, provide energy to the cell, and organelles called vacuoles store substances in the cell. Organelles allow eukaryotic cells to carry out more functions than prokaryotic cells can. This allows eukaryotic cells to have greater cell specificity than prokaryotic cells. Ribosomes, the cellular structures where proteins are made, are found in both eukaryotic and prokaryotic cells.

In some ways, a cell resembles a plastic bag full of Jell-O. Its basic structure is a cell membrane filled with cytoplasm. Like Jell-O containing mixed fruit, the cytoplasm of the cell also contains various structures, such as a nucleus and other organelles.

aryotic Cells

bacteria. Organisms with prokaryotic cells are called **prokaryotes**. They were the first type of organisms to evolve and are still the most abundant organisms today.

DID YOU KNOW?

Mature red blood cells (RBCs) do not have a nucleus present. RBCs contain hemoglobin protein that binds to oxygen and helps in the transport of oxygen throughout the body. In the absence of a nucleus, mature RBCs can accommodate more hemoglobin to carry oxygen.

Summary

- Prokaryotic cells are cells without a nucleus.
- Eukaryotic cells are cells that contain a nucleus.
- Eukaryotic cells have other organelles besides the nucleus.

Review

1. What is the cell nucleus?
2. What is the main difference between prokaryotic and eukaryotic cells?
3. Give an example of a prokaryotic cell.
4. Define organelle.
5. What is the advantage of having organelles?